

TEAM REBORN

MEMBERS NAME-

Apoorv Mishra
Abhinav apurv
Chakshu mehta
Aditi chaudhary

UNSTOP ID-apoorvmishra2007@gmail.com

The Corporate Heist

1. Summary

Our shortlist combines a historical performance model with explicit current-cycle policy rules. Two gradient-boosted regressors learn one-year performance from 20,000 historical hires: a fundamentals-only quality model and a diagnostic legacy model that also captures the older panel's preferences. The final score gives 85% of base-model weight to fundamentals, then applies the insider's debrief directly: strong nonlinear credit for sustained public-code contributions, a merit bonus for technically strong applicants from unseen institutes, and hard rejection for notice periods beyond 60 days or implausibly rapid executive promotion. Entity resolution ensures that one person appears at most once. High-confidence fabricated profiles are removed using consistency checks. The final output contains exactly 500 ranked test IDs.

2. Data cleaning and parsing

Field	Cleaning applied
Assessments	Converted technical values such as 0.84, 84/100 and 84% to 0-100; converted aptitude percentages to 0-10.
Experience	Converted Fresher, <1 year, month counts, 5+ yrs and >20 to numeric years; summed durations inside career_path.
Ratings and KPI	Mapped named rating bands to 1-5 and normalized Yes/Y/TRUE/1 and No/N/FALSE/0.
Compensation	Converted rupees, lakh/LPA, crore and dollar-like values to approximate lakh/year before calculating expected hike.
Notice period	Converted immediate, serving-notice, day and month formats to days; values above 60 fail the current-cycle rule.
Text and lists	Counted skills/certifications, extracted recruiter-note indicators and measured role-skill keyword coverage.

Examples found in the files include technical scores stored as both 0.43 and 43, experience stored as 63 months and 5+ yrs, and compensation stored as INR 29.1 lakh, 2,910,000 or 0.291 Cr. Missing values remain missing until median imputation inside the model; they are not silently converted into favourable zeros.

3. What drives a great hire - findings

- Technical assessment is the strongest stable signal. Dev winners average roughly 78 versus 55 for non-winners.
- Aptitude, recent rating and KPI attainment provide consistent additional separation.
- Career consistency matters: large differences between summed career durations and claimed experience are associated with substantially lower historical scores.
- Role fit is measured from applied role, skills, title and career history rather than degree labels alone.
- Historical institute, city and referral signals improve old-cycle validation but are treated as policy effects, not candidate quality.

Dev-set retrieval of 150 revealed winners

Quality		77/150
Legacy		94/150
Historical blend		91/150

Validation method. Each train model predicts the continuous `post_hire_score`. We rank all 2,999 dev applicants and count revealed winners in the first 150 positions. The fundamentals-only model finds 77 winners. The legacy model finds 94, and the historical blend finds 91. Random selection would retrieve about 7.5 winners on average. These results verify the parsing and ranking pipeline without treating the arbitrary sample submission as a label.

Stability checks

- All seeds are fixed at 42; repeated clean runs produce byte-identical `submission.csv` files.
- The complete clean run finishes in about 20 seconds on the build machine, comfortably inside the five-minute limit.
- The official checker reports OK; ranks are consecutive, IDs are unique and every ID exists in `test.csv`.

Why two models?

The legacy model is useful evidence that the historical process contained strong institution, location and channel effects. The debrief explicitly says most of those effects disappeared after the reorganisation. Therefore the final cycle score uses only 15% legacy-model weight and 85% fundamentals-model weight before applying current-cycle rules.

4. The current cycle

Debrief evidence	Implementation
Public code	No bonus below 10. Bonuses increase at 10, 20, 30, 50 and 100 contributions, with the largest increase beginning at dozens of contributions. Missing/not tracked is neutral.
Rapid titles	Directors, heads and VPs with under eight years are rejected. Implausibly fast lead/staff/manager progressions receive a strong penalty.
Join within two months	Notice period above 60 days is a hard rejection. Exactly 60 days remains eligible.
New institutes	An institute unseen in train and dev earns a bonus only with technical score at least 80 and meaningful role-skill fit; unseen status alone gives no credit.
Old preferences removed	General pedigree, metro, referral, degree and company-size effects are prevented from dominating by the 85/15 quality/legacy blend.

5. Profiles excluded, and why

- 208 test applicants have notice periods above 60 days and are ineligible.
- The test extract contains an obvious new cluster of executive titles reached with very low experience; these profiles are excluded under the committee's stated rule.
- Impossible graduation age, professional experience greater than feasible age, or multiple severe contradictions trigger a fabricated-profile rejection.
- Large career-duration gaps and high assessment combined with a rating of 1 contribute to the fabrication score; one weak anomaly alone is not enough.
- Normalized phone and e-mail identify repeated recruiter entries. Connected duplicates are ranked once, retaining the earliest file occurrence as the deterministic candidate ID.

Final shortlist audit: 500 unique candidate IDs, zero repeated normalized phone/e-mail values, zero notice-period violations and zero obvious low-experience executive-title violations.

Entity-resolution details

Phone values are reduced to their final 10 digits. E-mail local parts are lowercased, punctuation is removed and +jobs aliases are stripped. Rows connected by either canonical key form one entity. In the revealed ledger, every winner-containing duplicate group used the earlier occurrence, supporting the tie-break rule used for the Vault.

6. Model and selection procedure

- Engineer numeric consistency, assessment, performance, experience, compensation, text-indicator and role-fit features.
- One-hot encode the selected categorical groups after combining category vocabularies without using any outcomes from dev or test.
- Fit two HistGradientBoostingRegressor models to post_hire_score with 300 iterations, 31 leaves, learning rate 0.05 and L2 regularization 2.
- Blend 85% quality and 15% legacy predictions, then add debrief-derived code and unseen-institute bonuses.
- Apply hard policy/fabrication exclusions, collapse duplicate entities, sort descending and take exactly 500.

7. What did not work

A direct top-5% classifier trained from thresholded historical scores retrieved fewer dev winners than continuous-score regression. A first 20-feature CatBoost baseline retrieved 60/150 dev winners. Blindly using all historical categorical preferences improved old-cycle validation but conflicted with the debrief, so those signals were isolated in the legacy model and heavily down-weighted. Missing public-code values were not treated as zero because 'not tracked' does not prove inactivity.

8. External resources and AI tools

- No external datasets were used.
- Libraries: Python 3.11 standard library, NumPy, pandas and scikit-learn.
- ChatGPT/Codex was used for problem interpretation, exploratory analysis, code drafting, rule design, testing and documentation.
- Google Antigravity IDE was used by the team for local environment setup and execution guidance.

9. How to run

Step	Command / expected result
Place files	Put train.csv, dev.csv, dev_winners.csv and test.csv in the working directory beside the code folder.
Execute	python code/main.py
Output	submission.csv is written in the working directory with exactly 500 rows and columns rank,candidate_id.
Check	python check_format.py submission.csv test.csv
Runtime	Approximately 20 seconds on the build machine; random seed 42.